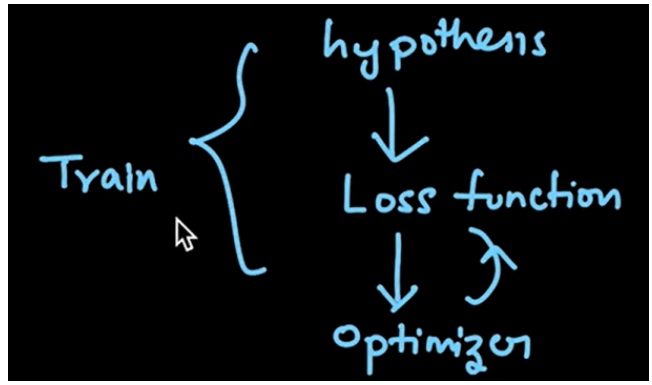
# Overfitting - Regularization

Tuesday, March 10, 2026 1:32 PM

ML model : train itself on input values X – So predicted Y values are close to the actual values as possible
– the difference b/w prd val and act val is error – error is used to evaluate the model

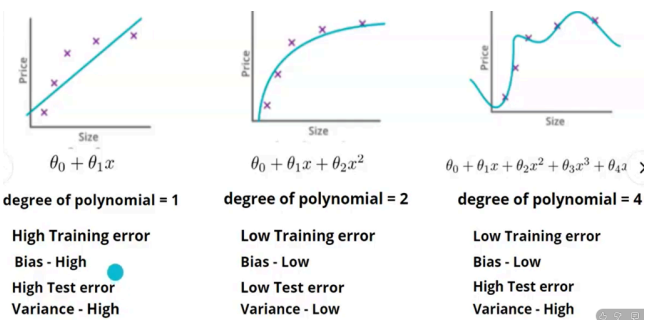



For supervised ML Model error has:

Bias error – model is not trained well with the training data

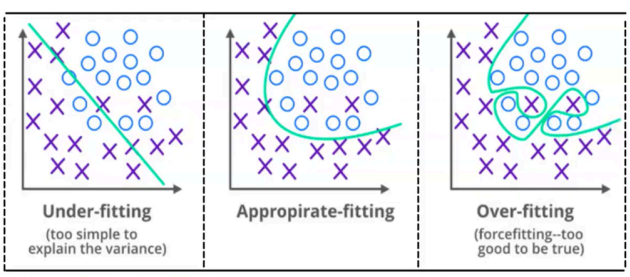
Variance error – training data has low error – but – New data comes out high error

Noise - Outliers

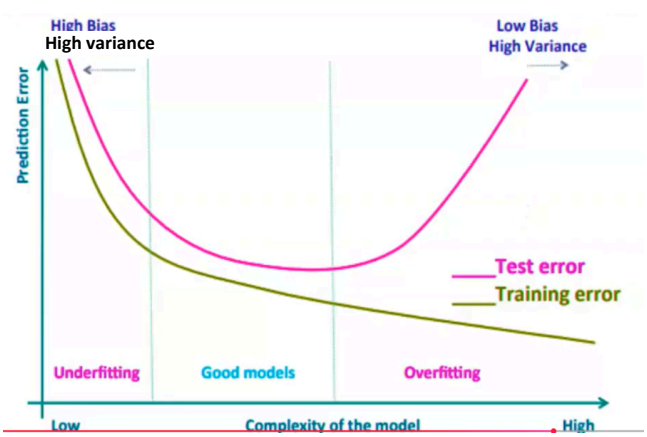



| Under fitting | Generalized model | Over fitting |
|---|---|---|

## Classification




The are two classes above x and o




-variance and bias are inversely proportional
-Bias and variance trade off - find a good point in common to maintain the bias and variance to avoid model underfit and overfit

How to reduce Overfitting:
-increase training data – could be Synthetic data
-early stopping before the error
-ridge and lasso regularization
? Dropout for neural networks – dropout few neural nodes-reduce model complexity (I.e.) degree of polynomial

Regularization: to prevent overfitting:

Ridge reg (L2): loss = (1/n Ey-y^) +lambda o2+.....
Try to reduce both A (init loss value)and B(lambda) to minimum

Lasso L2: init loss +lamda|o|+.....

Same set both A and B to very small values
elastic net : combination lasso and redge

ML -
Ols model check all lasso ridge elastic net are same

Polynomial
This is where the parabola and curves and other degree of curves come in play to pass our line of regression through maximum data points.
This is all just to get that best fit line to pass through the majority of the data points.